Portfolio - Individual

AI-Systems Minor

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1 Student Profile

My passion for technology started loong before I began my studies at Fontys University of Applied Sciences. From assembling computers to troubleshooting them, I was always trying to learn more and more about how computers work. This curiosity rapidly turned into a broader interest in the digital world and the many possibilities it offers. Once I joined this university, I discovered a whole new dimension of computing: the field of artificial intelligence and data mining. What I found most interesting is how giving the computer a dataset, with some proper training, we can see all sorts of predictions and correlations, most of the time them being accurate.

In the past years, artificial intelligence has gained significant traction and attention from people who have integrated it into their daily routine, from using it to get a recipe for a quick meal to helping them in school assignments. But one task that AI has made considerably easier, is working with large amounts of data. In the past we would have to manually look over the data, analize and recognise patterns and then after come up with a conclusion. This all changed when the process became automated. For this reason, I decided to make use of this technology to come up with an application that can analyze a person's face and detect what is his mood based on facial expression. I think this application could have an impact across multiple domains, such as well-being monitoring, online recommendation systems and online learning.

1.1 Self reflection

During my studies at Fontys University, I've studied many topics which focus on programming, algorithms and data structures. During the 4th semester, I discovered a course that captivated my full attention from the first lesson, that being Data Mining. We discussed many topics like how we should handle large sets of data, many different regression models and neural networks. In this course we also had to do weekly assignments. My favorite one was an exercise where we had to predict house prices based on some features such as number of rooms, square feet and if it has a garade or not using a suited regression model. This couse sparked my curiosity in this domain and thats why I chose this minor.

1.2 SWOT Analysis

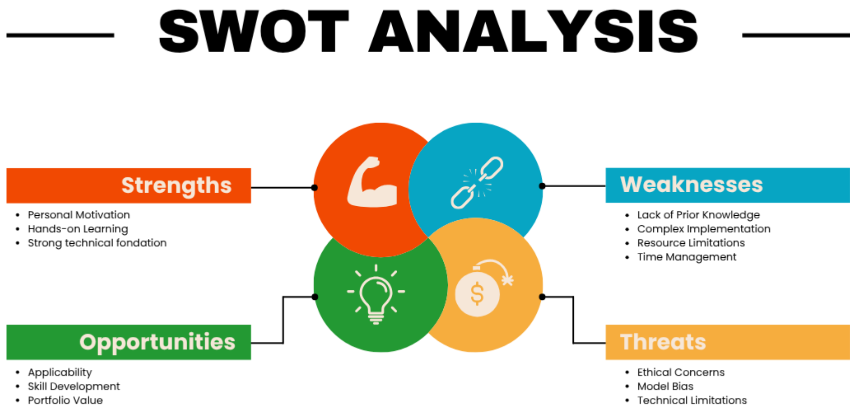

Figure 1: SWOT Analysis

Strengths

The project builds on my personal motivation and curiosity for AI, giving me a strong drive to complete it successfully. It also offers a hands-on learning experience where I can apply my technical knowledge in a practical context. My growing foundation in machine learning and computer vision supports the technical development of the model.

Weaknesses

Because I am still gaining experience in AI and computer vision, the project comes with a learning curve. The implementation can be complex, and limited time or computing resources may slow progress. Managing these aspects effectively will be essential to stay on schedule and maintain project quality.

Opportunities

This project allows me to explore a field with real-world applications, from well-being monitoring to adaptive learning systems. It provides the opportunity to strengthen my technical skills, learn new tools, and create something that can be showcased in my professional portfolio.

Threats

Potential risks include ethical concerns related to privacy and fairness, as well as the possibility of model bias if the dataset lacks diversity. Technical limitations such as lighting, camera quality, or real-time performance may also affect the system's accuracy.

2 Personal Learning Objectives

2.1 Learn about computer vision and machine learning

One of my main objectives for this project is to improve my ability to work with computer vision models, not just by studying their theory but by applying them to real-world image data. At the moment, I have some understanding of concepts such as convolutional neural networks and transfer learning, but I do not fully grasp how to implement them effectively for a task like mood detection. Through this project, I want to explore how to prepare and preprocess facial expression datasets, how to train and fine-tune different models, and how to evaluate their performance across various scenarios. By experimenting with multiple approaches and comparing their results, I aim to gain a more concrete understanding of how computer vision techniques can be used to build reliable systems that interpret human emotions.

2.2 Reflect on ethics of AI

During this project, I am hoping to expand my understanding of the implications of AI systems, primarily on the implications of AI systems on the recognition of facial emotion. I know some of the ethical limitations of AI, such as biases and privacy issues, but I have not had the opportunity to think on how they are relevant to my work. I hope to identify some of the risks and limitations concerning the fairness and accuracy of the system on different users and across multiple situations. This will be the first step in making my approach responsible, deriving a balanced and realistic understanding that recognizes the technical side of a project and the ethical concerns that must be considered. I hope to advance my work in a more responsible way within the social context

2.3 Managing study time

One problem that has frequently appeared during my studies, is that I tend to manage my time poorly and sometimes barely have any time to study. I tend to underestimate the workload and find myself in a difficult situation when there are deadlines involved. Although I am motivated, this habit tends to affect the way I work and my productivity. In this project I want to adopt a new strategy, by setting weekly deadlines and reserving 1-2 hours of self study per day. In my opinion this will help me set realistic milestones and prevent me from procrastinating and doing all the work in one day. This will not only help me sucessfully complete this project, but will also prove an useful skill in further academic and professional work.

3 Context

The inspiration for this project comes from a childhood curiosity. Since I was a little kid, I always was fascinated by Apple's products and how they were two steps ahead of their competition. In September 2017, when they introduced the IPhone X, they also announced a new feature that amazed and still amazes me to this day. That feature was facial recognition, the possibility to unlock your phone just by looking at it. That moment sparked my curiosity about how machines can recognize and interpret human faces. At first, I was impressed by the security aspect of unlocking a device with facial recognition, but later I started wondering whether similar technology could also understand more subtle aspects, such as a person's mood or emotional state. This thought stayed with me, and as I learned more about artificial intelligence during my studies, I realized that computer vision and machine learning could make this possible. My project idea grew out of this interest: if a phone can identify who you are, perhaps a system can also detect how you feel, and this could open the door to meaningful applications in areas like well-being, online learning, and personalized recommendations.

4 Theoretical Learning Objectives

Up to this point, I have little knowledge about machine learning and computer vision. To be able to build a reliable model that can recognize emotions in real time, I still need to build up my knowledge in these areas and practice by working on large datasets available on the internet.

4.1 Main Research Question

How effectively can computer vision and machine learning techniques detect and classify a person's mood based on facial expressions captured from images, videos, or real-time camera input?

4.2 Sub Research Questions

1. How does the ethnicity of a person affect the performance and fairness of mood detection models?
2. How does real-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability?
3. What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?

4. How does the model's accuracy in mood detection vary across different emotion categories?

A (sub)question is considered complete once I have gathered sufficient results, analysis, or reflection to confidently answer it based on evidence from my research and development process.

4.3 Topics

In this project, I plan to explore the following areas:

1. **Computer Vision:** I will learn how to detect and analyze faces from images and videos to recognize facial expressions.
2. **Machine Learning:** I will use machine learning models, such as convolutional neural networks, to classify emotions based on facial expressions.
3. **Datasets and Data Handling:** I will work with existing facial expression datasets and prepare them for training by cleaning, organizing, and analyzing their quality.
4. **Fairness and Ethics:** I will study how different factors, such as ethnicity, can affect the model's accuracy and how to ensure the system is fair and respects privacy rules.

4.4 Bloom's Taxonomy

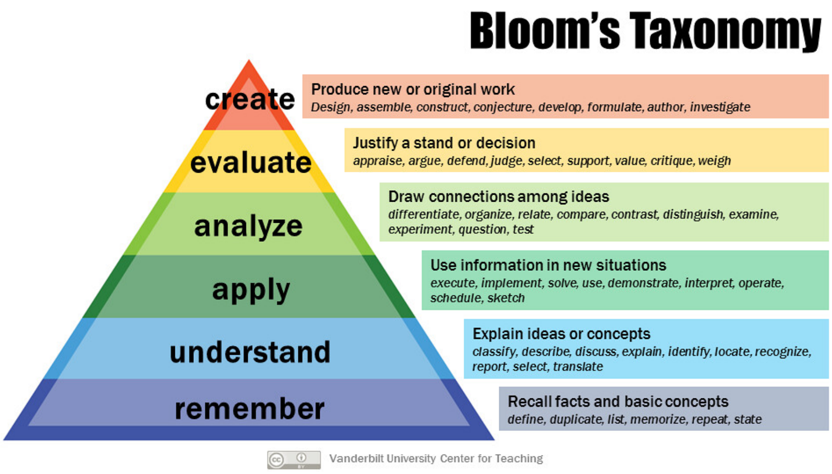

Figure 2: Bloom's Taxonomy

Computer Vision – Apply/Understand

By applying and understanding computer vision principles, I can learn how images are processed and how various algorithms detect and interpret facial features.

Machine Learning – Create/Analyze
This objective is a combination between Create and Evaluate. I aim to apply machine learning techniques to classfiy moods and then analyze how different models perform on specific datasets. By comparing their accuracy and limitations, I can determine which approach is more suitable for emotion detection.

Datasets and Data Handling – Analyze/Create
This objective goes hand in hand with Analyze and Create. By analyzing and preparing datasets to ensure reliability, I can determine if the available datasets on the internet are suitable for training mood detection models. This involves examining data quality and recognizing potential biases that may influence the results. After training the AI on datasets from the internet, I can create my own datasets by taking pictures of my friends expressing different moods and annotating them.

Fairness and Ethics – Evaluate
This objective focuses on evaluating the fairness and ethical implications of facial emotion recognition. I want to explore how differences in ethnicity may influence the accuracy and reliability of the model, and investigate potential biases that could arise from unbalanced datasets. In addition, I plan to study existing guidelines and policies for responsible AI development, so I can reflect on the standards that should be followed to ensure fairness, inclusivity, and ethical use of mood detection systems.

5 Learning Strategy

To be able to answer the research questions and reach my proposed learning objectives, I will apply several learning strategies:

1. **Desk Research**: Recap the lessons from the Data Mining course (semester 4) to strengthen my knowledge of machine learning and neural networks.

2. **Online Resources**: Study articles and watch YouTube tutorials on facial recognition and computer vision to support setting up my model.

3. **Fontys Lessons**: Attend the Deep Learning and Computer Vision lectures and prepare questions to ask lecturers afterwards.

4. **Development**: Experiment with different datasets and models to gain hands-on experience and develop the desktop app for presenting emotions in a user-friendly way.

Sub-Research Question	Strategy (including Learning Strategy)
How does the ethnicity of a person affect the performance and fairness of mood detection models?	Evaluate model performance across diverse datasets, analyze potential biases, and compare accuracy results between different ethnic groups. **Learning Strategy:** Desk Research (study bias and fairness in AI), Development (experiment with diverse datasets).
How does real-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability?	Test the model on live webcam input and compare results with static images and pre-recorded videos to measure both accuracy and responsiveness. **Learning Strategy:** Development (implement experiments), Online Resources (tutorials on real-time computer vision).
What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?	Conduct desk research on AI ethics and legal frameworks (e.g., GDPR, EU AI Act) and reflect on responsible use, privacy, and fairness in application. **Learning Strategy:** Desk Research (policies and ethics), Fontys Lessons (ask questions on ethical AI).
How does the model's accuracy in mood detection vary across different emotion categories?	I will evaluate per-class accuracy using confusion matrices to identify which emotions are most accurately recognized and where misclassifications occur. **Learning Strategy:** Development (evaluate model performance using confusion matrices), Desk Research (study performance metrics and evaluation methods).

Table 1: Sub-Research Questions with Strategies and Learning Strategies

6 Theoretical Learning Objectives

The main question to answer in this project is as follows:
How effectively can computer vision and machine learning techniques detect and classify a person's mood based on facial expressions captured from images, videos, or real-time camera input?

To answer this question, first we need to answer the subquestions mentioned below:

6.1 First Sub Question:

How does the ethnicity of a person affect the performance and fairness of mood detection models?

Introduction to Algorithmic Bias

Computer vision models are not inherently neutral; they reflect the data they are trained on. In the context of facial emotion recognition (FER), the ethnicity of a subject significantly impacts model performance. If a dataset is unbalanced (containing mostly images of one demographic group), the model will "overfit" to the facial features, skin tones, and lighting conditions typical of that group, while failing to generalize to others. This phenomenon is known as Algorithmic Bias.

The Dataset Problem: FER2013 vs. Reality

The model developed in this project was initially trained on the FER2013 dataset. Research indicates that FER2013, like many early academic datasets, is heavily skewed towards White/Western faces. This can lead to a number of problems:

1. **Feature Overfitting:** The MobileNet model learns to detect emotions based on specific landmarks (e.g., the curve of a mouth, the crinkle of an eye). If the training data lacks diversity, the model may treat features common to specific ethnicities (such as different eye shapes or brow structures) as indicators of specific emotions rather than neutral physical traits.

2. **Performance Gap:** Consequently, when the model is tested on non-Western faces (e.g., Asian or Black individuals), the accuracy drops. The "recall" score (The ability to find all examples of an emotion) is significantly lower for minority groups because the model simply hasn't seen enough examples to recognize their expressions confidently.

The "Angry Black Man" Paradox

One of the most documented and harmful failures in facial recognition fairness is the "Angry Black Man" paradox. This occurs when a model disproportionately misclassifies "Neutral" or "Serious" expressions of Black men as "Angry."

- **Cause:** The model learns to associate darker skin tones or specific natural resting face structures with the label "Angry" because the few examples of Black individuals in the training set might have been biased towards negative emotions, or simply because the model confuses physical features with emotional signals.

- **Impact:** In a real-world deployment (e.g., an automated interview screener or a security system), this bias could lead to unfair penalization or profiling of Black individuals, flagging them as aggressive when they are simply neutral. This violates the fairness principle outlined in the EU AI Act “Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence”, 2021.

The "Polite Smile" Paradox

A significant paradox affects Asian demographics, often rooted in Cultural Display Rules rather than just physical features. This paradox occurs when models incorrectly classify expressions of embarrassment, anxiety, or politeness as "Happy."

- **Cause:** Western-centric datasets like FER2013 almost exclusively annotate the geometric feature of an upturned mouth as "Happiness." However, in many East Asian cultures (e.g., Japan, Thailand), smiling is frequently used as a social tool to mask negative emotions like embarrassment or confusion. Because the model lacks this cultural context, it sees the geometry of a smile and ignores other subtle cues (like the eyes) that might indicate distress.

- **Impact:** This bias is critical in healthcare or wellbeing applications. For example, an AI therapist or mental health monitoring app might incorrectly classify a distressed Asian user as "Happy" or "Doing well" simply because they are smiling out of polite discomfort. This leads to a false positive for well-being, potentially denying the user necessary help.

Mitigation Strategies and AffectNet

To address these ethnic disparities, simply training for more epochs is insufficient. The underlying data must be corrected:

1. **Dataset Balancing:** Moving from FER2013 to larger, like AffectNet provides a more balanced representation of global demographics, reducing the performance gap.

2. **Bias Auditing:** Running a "Bias Audit" (testing the model on a balanced dataset) is crucial to identifying and quantifying these risks before deployment.

Conclusion

Ethnicity plays a critical role in the reliability of mood detection models. Without intentional intervention, models trained on standard public datasets will exhibit significant bias, performing accurately on White faces while misinterpreting or under-performing on minority groups. Ensuring fairness requires a shift from model-centric optimization to data-centric curation, ensuring all ethnicities are represented equally in the training phase.

6.2 Second Sub Question:

How does reat-time camera input compare to static images or pre-recorded videos in terms of accuracy and usability

WORK IN PROGRESS

6.3 Third Sub Question:

What ethical considerations need to be taken into account when developing and applying facial emotion recognition systems?

6.3.1 Privacy and Data Protection

Facial data constitutes sensitive biometric information. To minimize risk, the project follows data minimization and purpose limitation, as required by the GDPR "General Data Protection Regulation (EU) 2016/679", 2016.

- **Local processing:** All inference is performed on-device; no raw images or identifiers are transmitted.
- **No permanent storage:** By default, images/frames are processed in memory only. Any diagnostic logging excludes personal data.
- **Data deletion:** Temporary buffers are cleared immediately after inference.

6.3.2 User Consent and Transparency

The application implements explicit, informed consent in compliance with Article 6(1)(a) of the GDPR "General Data Protection Regulation (EU) 2016/679", 2016.

- **Opt-in:** Camera access and emotion inference are disabled until the user enables them.
- **Just-in-time notices:** A short notice explains what is processed, for what purpose, and for how long.
- **Controls:** Clear on/off toggle, session indicator (e.g., icon/LED), and a link to the privacy summary.

6.3.3 Data Storage and Security

If short-term storage is required for evaluation, the design follows the principles of integrity and confidentiality from Article 5 of the GDPR "General Data Protection Regulation (EU) 2016/679", 2016 and ISO-style information security best practices.

- **Scope:** Store only what is strictly necessary (e.g., aggregated metrics).
- **Protection:** Encrypt at rest; restrict access to authorized personnel.
- **Retention:** Define retention windows and automatic deletion schedules.

6.3.4 AI Bias and Fairness

The model may inherit bias from training data, which is addressed in line with the fairness principles of the EU AI Act "Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence", 2021. Mitigations include:

- Inspecting class distribution and augmenting underrepresented classes.
- Reporting class-wise metrics and confusion matrices.
- Avoiding high-stakes use; keeping predictions informational, not determinative.

6.3.5 Ethical Use and Context

The main goal of this project is to use emotion detection in a positive and helpful way, such as reminding users to take breaks or encouraging them when they show positive emotions. The system is not made for surveillance, profiling, or influencing people's behaviour. All demonstrations and tests clearly explain the purpose of the system to anyone involved. Users have full control over when the application is active and can pause or stop it at any time. Any future version of the app will show a clear indicator when the camera is on and will include a short, easy-to-understand privacy explanation. By setting clear limits and responsible uses, the project aims to keep emotion detection ethical and focused on helping people, consistent with the IEEE's principles of human wellbeing and accountability "Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems", 2019.

6.3.6 Regulatory Alignment

The project aligns with key European data-protection and AI-ethics frameworks, including the General Data Protection Regulation (GDPR) "General Data Protection Regulation (EU) 2016/679", 2016 and the upcoming EU AI Act "Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence", 2021. Under GDPR, it respects all major principles: lawfulness, fairness, transparency, purpose limitation, data minimisation, and integrity/confidentiality. The design treats all facial analysis as sensitive processing and ensures that it stays within a low-risk category by design—local processing, explicit user consent, and transparent operation. In terms of governance, documentation is maintained for the model's purpose, assumptions, and limitations to promote accountability. The project also draws on the IEEE Ethically Aligned Design framework "Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems", 2019, focusing on human agency, accountability, and the promotion of wellbeing. Together, these measures ensure that the prototype not only complies with current legal requirements but also anticipates future standards for trustworthy and responsible AI systems.

Risk & Mitigation Summary

Risk	Description	Likelihood	Mitigation
Privacy leakage	Unintended storage/transmission of frames	Low	On-device processing, no raw logs, code review
Bias	Uneven accuracy across groups/classes	Medium	Class-balance checks, per-class metrics, iterative tuning
Misuse	Use for surveillance without consent	Low/Med	Usage policy, UI indicators, opt-in only
Security	Unauthorized access to temp data	Low	Encryption, least-privilege access, short retention

6.3.7 Conclusion

With local processing, explicit consent, limited storage, fairness checks, and clear scope limits, the prototype adheres to responsible AI practices and prepares for compliant, user-respecting experimentation. This approach demonstrates alignment with the ethical frameworks of GDPR "General Data Protection Regulation (EU) 2016/679", 2016, the EU AI Act "Artificial Intelligence Act: Proposal for a Regulation Laying Down Harmonised Rules on Artificial Intelligence", 2021, and IEEE Ethically Aligned Design "Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems", 2019.

6.4 Fourth Sub Question:

How does the model's accuracy in mood detection vary across different emotion categories?
To answer this question, three iterations of the model were evaluated:

1. **Model A:** 7 Classes (All emotions), trained for 25 Epochs.
2. **Model B:** 7 Classes (All emotions), trained for 35 Epochs.
3. **Model C:** 3 Classes (Positive, Negative, Neutral), trained for 35 Epochs.

The performance of each model was analysed using the per-class F1 scores and confusion matrices. These reveal how reliably the system can distinguish between specific emotions and which categories remain challenging for the MobileNet architecture trained on FER2013.

Model A: 7 Classes, 25 Epochs

Model A shows a clear imbalance in performance across emotion categories. Emotions with highly expressive and distinct facial cues perform best, while subtle or ambiguous categories remain problematic.

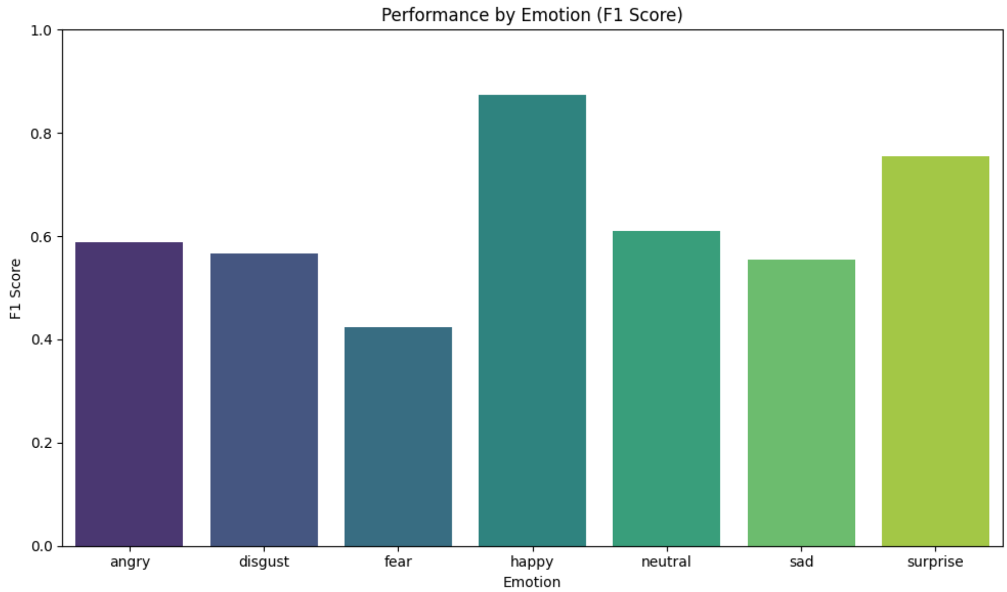

Figure 3: Model A — F1-score per class (7 classes, 25 epochs)

- **Highest-performing emotions:** *happy* (F1 ≈ 0.87) and *surprise* (F1 ≈ 0.76). These emotions have strong visual signals and are well-represented in FER2013.
- **Moderate performance:** *angry* (F1 ≈ 0.59), *neutral* (F1 ≈ 0.61), *disgust* (F1 ≈ 0.57), and *sad* (F1 ≈ 0.55). These emotions often overlap visually, leading to frequent misclassifications.
- **Lowest-performing emotion:** *fear* (F1 ≈ 0.42), which is heavily confused with *sad*, *surprise*, and *angry*. This aligns with the literature, as fear expressions are subtle and inconsistently depicted in FER2013.

The confusion matrix confirms broad dispersion among negative emotions, suggesting the model struggles to differentiate fine-grained emotional cues under limited training duration.

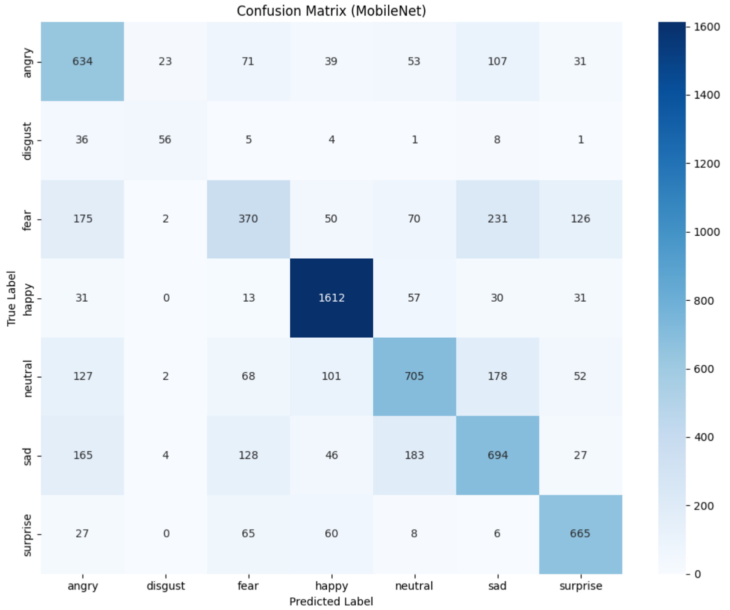

Figure 4: Model A — Confusion Matrix (7 classes, 25 epochs)

Model B: 7 Classes, 35 Epochs

Increasing the training time to 35 epochs improves overall stability and class separation.

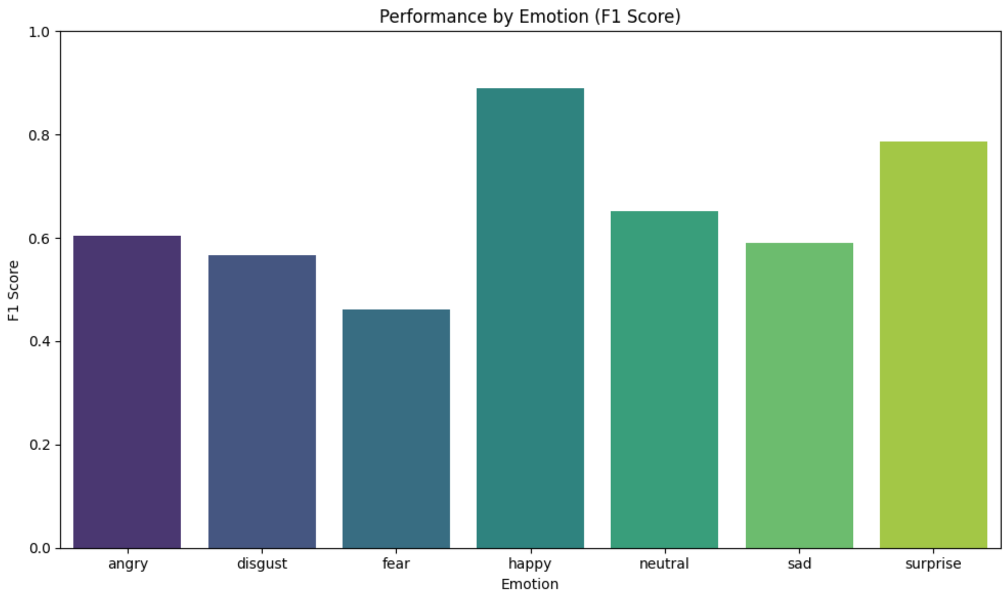

Figure 5: Model B — F1-score per class (7 classes, 35 epochs)

- **Improvements:** Most emotion categories show higher F1 scores compared to Model A. Notably, *happy* remains the strongest class (F1 $\approx$ 0.89), and *surprise* increases to roughly 0.79.

- **Negative emotion patterns:** *angry*, *disgust*, *sad*, and *neutral* all experience modest improvements, but the confusion matrix still reveals strong overlap among these categories.
- **Fear remains challenging:** Even with extended training, *fear* continues to have the lowest performance (F1 $\approx$ 0.46), confirming its status as the least distinguishable category.

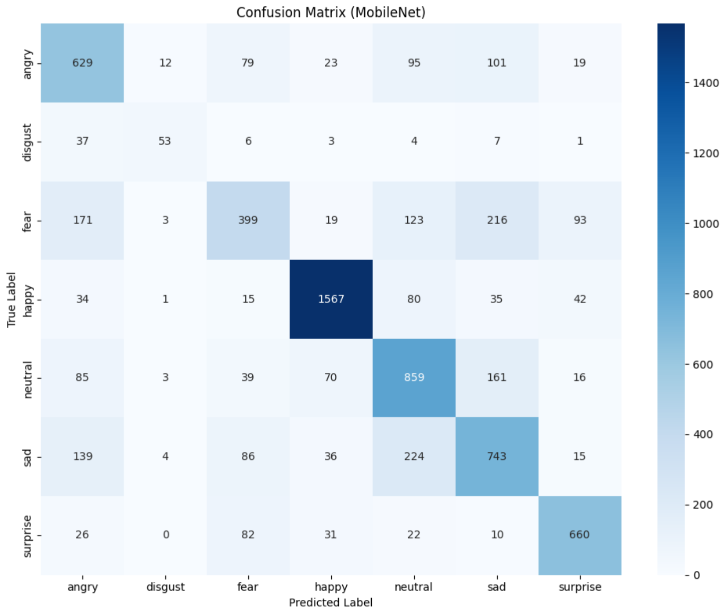

Figure 6: Model B — Confusion Matrix (7 classes, 35 epochs)

Overall, Model B demonstrates that longer training helps refine class boundaries, but the inherent ambiguity and dataset imbalance in FER2013 limit the achievable accuracy for certain emotions.

Model C: 3 Classes (Positive, Negative, Neutral), 35 Epochs

When emotions are grouped into broader categories, performance improves substantially. This setup reduces the complexity of distinguishing subtle expressions and mitigates dataset imbalance.

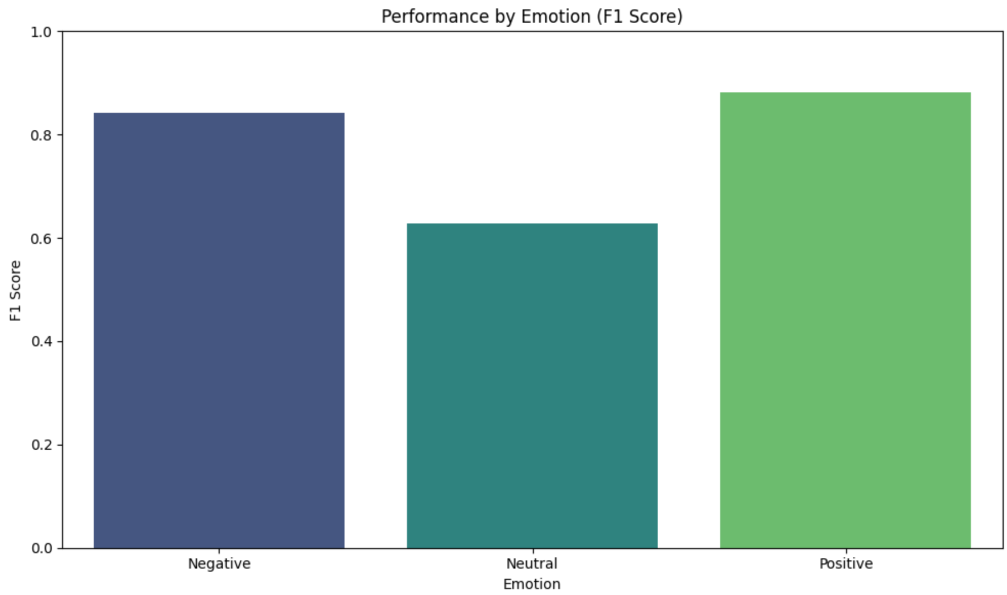

Figure 7: Model C — F1-score per class (3 classes, 35 epochs)

- **Positive:** The strongest category (F1 ≈ 0.88), driven largely by the clarity of the *happy* class.
- **Negative:** Strong performance (F1 ≈ 0.84), showing that grouping multiple negative emotions stabilises classification.
- **Neutral:** The weakest category (F1 ≈ 0.63), frequently misclassified as Positive or Negative. This reflects the inherent ambiguity of neutral expressions and their overlap with low-intensity emotions.

The confusion matrix shows that Neutral expressions often sit between Positive and Negative cues, explaining the reduced F1 score compared to the other two classes.

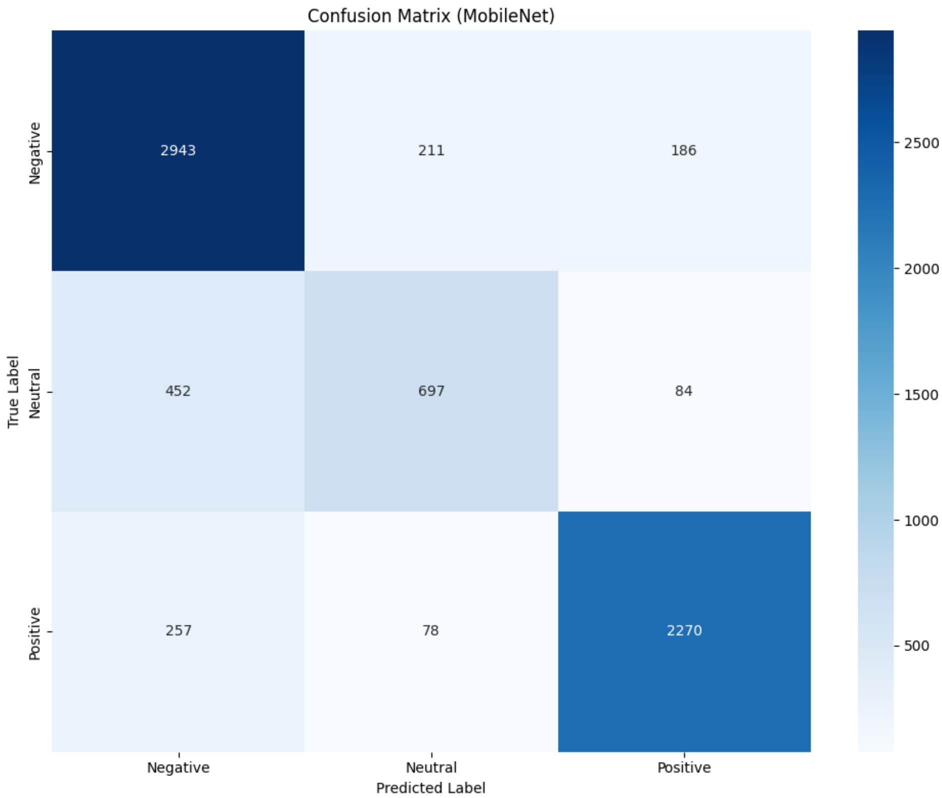

Figure 8: Model C — Confusion Matrix (3 classes, 35 epochs)

Conclusion

Across all three models, certain trends are consistent:

- Emotions with clear, distinct facial features (e.g., *happy*, *surprise*) consistently achieve the highest accuracy.
- Subtle, low-intensity, or culturally variable emotions (*fear*, *disgust*, *sad*) remain challenging regardless of training duration.
- Increasing epochs improves performance but does not fully resolve dataset-driven limitations.
- Reducing the classification task to broader emotion groups significantly increases accuracy, especially for Positive and Negative categories.

In summary, the model's accuracy varies substantially across emotion types, with high performance on expressive emotions and weaker performance on subtle or ambiguous categories. This variation reflects both the limitations of FER2013 and the biological reality that some emotions are inherently harder to recognise.

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